

Elizaveta (Liza) Rebrova

<https://erebrova.github.io/>

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Academic employment

2026–present	University of Oxford <i>Associate Professor, Mathematical Institute</i> <i>Senior Research Fellow, New College</i>
2021–2026	Princeton University, ORFE Department <i>Assistant Professor</i> Affiliated with the Program in Applied and Computational Mathematics (PACM) and the Center for Statistics and Machine Learning (CSML)
2021	Lawrence Berkeley National Laboratory <i>Postdoctoral Scholar, Tensor Factorization in Machine Learning</i>
2018–2020	University of California, Los Angeles, Department of Mathematics <i>Assistant Adjunct Professor</i> ; postdoc mentors: D. Needell, T. Tao
2017	Lawrence Berkeley National Laboratory Graduate Student Intern. Mentor: X. S. Li

Education

2013–2018	Ph.D. in Mathematics, University of Michigan, Ann Arbor Thesis: <i>Spectral properties of heavy-tailed random matrices.</i> Advisor: Roman Vershynin
2007–2012	Specialist in Mathematics, Lomonosov Moscow State University B.S.+M.S. equivalent; diploma with honors. Thesis: <i>Sobolev classes on infinite dimensional spaces.</i> Advisor: Vladimir I. Bogachev
2011–2012	Yandex School of Data Analysis, Moscow M.S. equivalent in Computer Science

Grants and selected awards

2023–2026	NSF DMS-2309685 , PI. “Outliers are not what they seem: data-aware, flexible, and robust randomized iterative methods”
2021–2024	NSF DMS-2108479 , Co-PI. “Fast, Low-Memory Embeddings for Tensor Data with Applications” (PI: D. Needell; collaborative partner: M. Iwen)
2018–2020	Capital Fund Management sponsored postdoc grant, UCLA
2017, 2018	Rackham Dissertation Fellowship (2017); Allen Shields Memorial Fellowship (2018), University of Michigan; NSF ORISE scholarship for LBNL internship (2017)

Advising and mentoring

Ph.D. students	Current: Sofia Shvaiko. Graduated: Jackie Lok (2026); Shambhavi Suryanarayanan (2026); Abraar Chaudhry (2024, co-advised with A. A. Ahmadi).
Master’s thesis	Nicolo Grometto (2023), <i>Essays on Random Matrices.</i>
Undergraduate	Senior theses and independent research at Princeton (2021–2026); five students in UCLA’s CAM REU (2020).

Selected publications and preprints

- 2026 J. Haddock, A. Ma, E. Rebrova. *Quantile-based loss filtering for outlier-robust stochastic gradient descent*. Preprint, arXiv:2609.13040.
- 2026 X. Du, E. Rebrova, M. Dereziński, S. Na. *Inference for Newton methods with accelerated sketch-and-project via random scaling*. Preprint, arXiv:2609.12421.
- 2026 S. Suryanarayanan, E. Rebrova. *On trimming tensor-structured measurements and efficient low-rank tensor recovery*. **Linear Algebra and its Applications**, 750, 163–204.
- 2026 J. Lok, E. Rebrova. *Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations*. **SIAM Journal on Matrix Analysis and Applications**, 47(3), 1495–1529.
- 2026 A. Chaudhry, E. Rebrova. *Learning nonnegative matrix factorizations from compressed data*. **SIAM Journal on Matrix Analysis and Applications**, 47(3), 1277–1303.
- 2025 M. Dereziński, D. Needell, E. Rebrova, J. Yang. *Randomized Kaczmarz methods with beyond-Krylov convergence*. **SIAM Journal on Matrix Analysis and Applications**, 46(4), 2558–2588.
- 2025 M. Dereziński, D. LeJeune, D. Needell, E. Rebrova. *Fine-grained analysis and faster algorithms for iteratively solving linear systems*. **Journal of Machine Learning Research**, 26(144), 1–49.
- 2025 H. Jeong, D. Needell, E. Rebrova. *Stochastic gradient descent for streaming linear and rectified linear systems with adversarial corruptions*. **SIAM Journal on Mathematics of Data Science**, 7(2), 516–541.
- 2025 J. Lok, R. Sonthalia, E. Rebrova. *Error dynamics of mini-batch gradient descent with random reshuffling for least squares regression*. **Algorithmic Learning Theory (ALT)**, PMLR 272, 736–770.
- 2024 M. Dereziński, E. Rebrova. *Sharp analysis of sketch-and-project methods via a connection to randomized singular value decomposition*. **SIAM Journal on Mathematics of Data Science**, 6(1), 127–153.
- 2022 J. Haddock, D. Needell, E. Rebrova, W. Swartworth. *Quantile-based iterative methods for corrupted systems of linear equations*. **SIAM Journal on Matrix Analysis and Applications**, 43(2), 605–637.
- 2021 E. Rebrova, D. Needell. *On block Gaussian sketching for the Kaczmarz method*. **Numerical Algorithms**, 86(1), 443–473.
- 2021 M. A. Iwen, D. Needell, E. Rebrova, A. Zare. *Lower memory oblivious (tensor) subspace embeddings with fewer random bits: modewise methods for least squares*. **SIAM Journal on Matrix Analysis and Applications**, 42(1), 376–416.
- 2018 E. Rebrova, K. Tikhomirov. *Coverings of random ellipsoids, and invertibility of matrices with i.i.d. heavy-tailed entries*. **Israel Journal of Mathematics**, 227(2), 507–544.
- 2018 E. Rebrova, R. Vershynin. *Norms of random matrices: local and global problems*. **Advances in Mathematics**, 324, 40–83.

Professional service and leadership

Editorial	Associate Editor, <i>Information and Inference: A Journal of the IMA</i>
Organisation	Co-organiser, Numerical Linear Algebra workshop, FoCM 2026, Vienna, 16–18 July, with Vanni Noferini and Françoise Tisseur. Minisymposia: SIAM Annual Meeting (2025); SIAM MDS, SIAM Linear Algebra, CISS (2024); SAMPTA (2023); ICCOPT (2022).
Committees	Princeton Day of Statistics (2021, 2023); Women and Mathematics programme in Mathematical Data Science, IAS (2022; also application review panel)
Reviewing	NSF panels (2021, 2022, 2025, 2026). Journals include SIMAX, SISC, SIAM Journal on Imaging Sciences, SIAM Journal on Optimization, LAA, JMLR, Bernoulli, Foundations of Computational Mathematics, Mathematics of Computation, Numerische Mathematik, Numerical Algorithms, BIT Numerical Mathematics, and Information and Inference.

Teaching experience

2021–2026	Princeton: Probability and Stochastic Processes; Convex and Conic Optimization (both graduate); Networks
2018–2020	UCLA: Probability; Statistics; Stochastic Processes; Optimization
2013–2017	Michigan: Calculus I; Differential Equations; Computational Finance; Math Lab tutoring
2008–2013	School 57 and Kolmogorov high school, Moscow: calculus and algebra

Selected invited talks and lectures

July 2026	FoCM, Random Matrices workshop, Vienna. “Structure-aware concentration for modewise tensor sketches”
June 2026	RandNLA workshop, EPFL, Lausanne. “On regularization via early stopping for least squares regression”
December 2025	Applied Mathematics Seminar, Virginia Tech
November 2025	Applied and Computational Mathematics Seminar, Temple University
May 2025	Electrical and Computer Engineering Seminar, University of Delaware
March 2025	SIAM Computational Science and Engineering conference, Fort Worth; Stochastics Seminar, Georgia Tech
October 2024	SIAM Mathematics of Data Science, Atlanta; minisymposium “Structure in Data: Theory, Learning, and Algorithms”
June 2024	Five lectures, Graduate Summer School on Random Matrices and Applications, University of Michigan, Ann Arbor
May 2024	SIAM Linear Algebra, Sorbonne University, Paris; minisymposium “Recent Advances and Critical Challenges in Randomized Linear Algebra”
September 2023	Probability Seminar, Brown University
June 2023	FoCM, Paris: Random Matrices and Theoretical Foundations of Data Science workshops
November 2022	High-dimensional Statistics and Probability Seminar, Harvard University